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To cite this article: Ruoyan Jin *et al* 2026 *Mach. Learn.: Sci. Technol.* **7** 015012

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OPEN ACCESS

RECEIVED
5 September 2025REVISED
3 December 2025ACCEPTED FOR PUBLICATION
19 December 2025PUBLISHED
14 January 2026

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Data-efficient machine-learning interatomic potential for studying radiation effects in germanium

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^{*} Author to whom any correspondence should be addressed.E-mail: ruoyan.jin@aalto.fi**Keywords:** machine-learning interatomic potential, radiation damage, germaniumSupplementary material for this article is available [online](#)

Abstract

Exposure to harsh radiation environments leads to displacement damage in semiconductor materials, ultimately degrading the device performance. Molecular dynamics (MD) is a powerful method for simulating the dynamic processes of radiation-induced defect generation, clustering, and evolution, which are often beyond the reach of experimental observation. In this work, we introduce a machine-learning Gaussian approximation potential for germanium, specifically designed to describe properties related to radiation-induced collision cascades and the resulting damage. A repulsive potential is incorporated to accurately capture short-range interactions during cascade simulations. In addition, we reproduce elastic, thermal, and vibrational properties, and accurately capture the energetics of vacancies and self-interstitials. These high-fidelity predictions are comparable to *ab initio* calculations while requiring only a compact training database. This potential is expected to facilitate accurate MD simulations of radiation damage in germanium with greatly improved accuracy compared to empirical potentials.

1. Introduction

High-purity germanium (HPGe) detectors are the preferred choice for high-resolution gamma-ray spectrometers [1], x-ray detectors [2] and dark matter searches [3], due to their superior energy resolution and efficiency. As a semiconductor detector designed for the detection of ionizing radiation, HPGe detectors are vulnerable to damage from charged particles or photons, which can affect their performance [4]. The dynamic process of defect production is extremely fast and far from equilibrium, making experimental studies nearly impossible. However, atomistic simulations provide powerful methods for exploring the generation, evolution, and clustering of radiation-induced defects [5]. Quantum mechanical methods (QM), such as density functional theory (DFT), provide a relatively accurate Born-Oppenheimer potential energy surface (PES) and can be used to determine the structure of radiation-induced defects, defect formation energies, migration energy barriers [6], and threshold displacement energies [7, 8], which are fundamental quantities for characterizing the primary state of radiation damage. However, due to the limitations in simulation length and timescale, relying solely on QM simulations to fully understand radiation damage is computationally infeasible. As a result, modern molecular dynamics (MD) simulations have become a crucial tool for modeling radiation effects [9].

In MD simulations, a key input is an interatomic potential (IP) or force field, which serves to reproduce the PES. Simulating radiation damage is a challenging task for IPs, as the material undergoes numerous atomic-scale changes during cascade evolution. These processes include many-body atomic collisions, localized melting, rapid recrystallization under extreme temperature and pressure gradients, and clustering of defects [10]. It is difficult for empirical potentials to accurately capture all these properties, and the simulation results may vary depending on the chosen IP [5]. Although several empirical

potentials for Ge are well established, including the Tersoff [11], Stillinger–Weber [12], and MEAM [13] formalisms, they exhibit notable discrepancies in describing the formation, interaction, and evolution of irradiation-induced defects. Significant differences have been reported in predicting defect dynamics [14, 15], melting temperature [16], and the number of surviving defects [17]. These findings underscore the need for a potential that can reliably model radiation-induced damage in Ge.

Over the past decade, machine learning-based IPs (MLIPs) have been widely employed in MD simulations to study various material properties with improved accuracy [18, 19]. By learning the PES from a database generated by QM calculations [20–22], machine learning models can capture complex atomic interactions directly from the data, offering a promising approach to overcome the limitations of empirical potentials. In 2018, Bartók *et al* [23] proposed a general-purpose MLIP for silicon based on the Gaussian approximation potential (GAP) framework [20, 24], demonstrating the feasibility of developing a MLIP capable of comprehensively describing materials on the atomic scale. Since then, this kernel-based MLIP framework has been successfully employed to investigate radiation damage in tungsten [10] and silicon [25, 26], demonstrating the applicability of GAP in radiation damage studies. In parallel, MLIPs based on alternative frameworks have been developed for radiation damage studies in various materials [27–29], and applied to investigate extended defects [30] and mechanical properties [31]. These advances demonstrate the promise of employing MLIPs in this field. Here, we leverage this approach to investigate the physical properties related radiation damage in Ge.

In this study, we use the GAP framework [20, 24] to develop a data-efficient MLIP to investigate radiation damage in Ge. The GAP framework is selected because it fits naturally into our in-house ecosystem for accurate and efficient radiation-damage simulations using GAP-based MLIPs. At the core of this ecosystem is the TurboGAP [32, 33] code, a versatile MD engine specifically designed for GAP formalism. Key modules that are essential for reliable radiation-damage simulations have been recently implemented in TurboGAP [34].

The GAP formalism offers accuracy comparable to newer MLIP frameworks such as the moment tensor potential (MTP) [21] and atomic cluster expansion (ACE) [35]. Comparable accuracy between GAP and MTP for elemental systems (including Ge) and for binary compounds has been reported in [36] and [37], respectively. Likewise, [37] and [38] show that GAP achieves accuracy similar to or even higher than ACE. In the context of radiation damage, a comparable level of accuracy between GAP and MTP in predicting defect dynamics is demonstrated in [36, 39, 40]. Also, a similar performance between GAP and Neuroevolution potentials (NEP) [41] is reported in [28]. Tabulated-GAP (TabGAP) [42] is a more computationally efficient variant of GAP; however, it is primarily developed for metals [43] and provides significantly lower accuracy for covalently bonded materials and the complex atomic environments typical of displacement cascades.

GAP models using the smooth overlap of atomic positions (SOAP) descriptor [44] are slower than some more recent frameworks. To address this limitation, we use `soap_turbo` many-body descriptor [33], an optimized variant of SOAP descriptor that delivers faster simulations and improved accuracy [33, 45]. Using this framework, we recently carried out cascade simulations in silicon with primary knock-on atom (PKA) energies up to 10 keV in systems containing up to one million atoms [34].

In terms of data efficiency, GAP performs on par with most modern MLIP frameworks [37]. This is a critical advantage for radiation-damage simulations, where a representative database might include crystalline, defected, liquid, and amorphous configurations spanning a wide variety of atomic environments. A data-efficient framework therefore plays an important role in enabling a compact yet comprehensive database suitable for radiation-damage applications. Another key feature of GAP for radiation-damage studies is its built-in mechanism for stiffening the potential. Proper augmentation of a repulsive potential is essential for a MLIP to perform reliably in high-energy collision simulations.

This model is trained on a compact yet diverse dataset comprising 451 configurations, carefully selected to capture key aspects related to radiation damage. The potentials exhibits strong performance in describing short-range interatomic interactions, predicting defect properties, and capturing the melting point. It also shows excellent agreement with DFT reference results for elastic, thermal, and vibrational properties.

2. Method

2.1. Potential details

The GAP employs the Gaussian process regression framework, which has been extensively discussed in [20, 46]. Here, the key formalism and hyperparameter settings are presented. In GAP, the potential

energy E_i of an atom i is expressed in the following sum:

$$E_i = \delta^2 \sum_{s=1}^M c_s K(\mathbf{q}_i, \mathbf{q}_s), \quad (1)$$

where the hyperparameter M denotes the number of descriptor points characterizing the selected representative environments from the training database. The hyperparameter δ defines the energy scale and can be adjusted for different types of descriptors. K is a kernel function that quantifies the similarity between two atomic environments, represented by the descriptors $\mathbf{q}_i$ for the target atom i and $\mathbf{q}_s$ for the reference atom s in the training database. The c_s are the fitting parameters [46]. During the fitting process, the regularization hyperparameters (σ_{energy} , σ_{force} and σ_{virial}), which can be specified for each configuration or atom, are used to encode the prior information regarding the accuracy of target values (energies, forces, and virial stresses) [47].

In this work, the GAP is trained with a simple two-body descriptor and an optimized SOAPs many-body descriptor, `soap_turbo` [33]. The two-body descriptor, based on the interatomic distance, captures pairwise energy contributions, while the many-body descriptor accounts for the complex interactions by incorporating bond angles and symmetrized bond distances [46]. The `soap_turbo` descriptor ensures permutational, translational, and rotational invariance [33]. In addition to the GAP, a pair potential V_{pair} is employed to handle the steep short-range repulsion, where the energy scales are difficult for a single kernel to capture [23]. The total energy E_{tot} of a system containing N atoms is expressed by the following expression [10]:

$$\begin{aligned} E_{\text{tot}} &= \sum_{i < j}^N V_{\text{pair}}(r_{ij}) + \sum_i^N E_{\text{GAP}} \\ &= \sum_{i < j}^N V_{\text{pair}}(r_{ij}) + \delta_{2b}^2 \sum_i^N \sum_{s=1}^{M_{2b}} c_{s,2b} K_{2b}(\mathbf{q}_{i,2b}, \mathbf{q}_{s,2b}) + \delta_{mb}^2 \sum_i^N \sum_{s=1}^{M_{mb}} c_{s,mb} K_{mb}(\mathbf{q}_{i,mb}, \mathbf{q}_{s,mb}), \end{aligned} \quad (2)$$

where the pair potential V_{pair} is calculated based on the distance r_{ij} between atom i and atom j , with $2b$ and mb distinguishing the two-body and many-body descriptors, respectively.

The hyperparameters used for these two descriptors, along with brief descriptions for each hyperparameter, are listed in table 1. We test several cutoff radii ranging from 5 to 8 Å and find that increasing the cutoff radius does not significantly improve the GAP's performance, but it substantially increases the training time. A cutoff r_{hard} of 5 Å is found to be the optimal balance between accuracy and computational cost, and the transition width is set to 0.5 Å. The δ value for the many-body descriptor is set much smaller than that of the two-body descriptor, causing the two-body descriptor to have a greater contribution to the total energy. For the remaining hyperparameters, $\sigma_{0,r}$, $\sigma_{0,\perp}$, l_{max} , α_{max} , ζ and the sparse method, we find that our GAP is insensitive to the changes in them, so we set them according to [33, 46].

2.2. DFT calculations

The energies, forces, and virial stresses of all training structures are calculated using VASP [48–50], with the generalized gradient approximation (GGA) exchange-correlation functional proposed by Perdew, Burke, and Ernzerhof (PBE) [51]. The core and 14 valence electrons ($3d^{10}4s^24p^2$) are described using the projector-augmented wave (PAW) method [52, 53] (the *Ge_d* PAW potential in VASP 6.4.3). The energy cutoff is 500 eV and a Gamma-centered mesh with k-spacing of $0.15 \text{ \AA}^{-1}$ is generated to sample the Brillouin zone. The energy convergence criterion is 10^{-7} eV, and the Gaussian-smearing method with a smearing width of 0.05 eV is employed to help convergence.

The convergence thresholds for both geometry optimization and the nudged elastic band (NEB) method [54] are set to $0.03 \text{ eV \AA}^{-1}$ for force and 10^{-5} eV for energy. The damped MD algorithm is employed to find optimized lattice vectors and atom positions. Other optimization parameters are set similar to DFT static calculations. The elastic constants and the bulk moduli are calculated using the VASPKIT code [55].

2.3. MD simulations

The GAP is trained using QUIP [56]. All MD simulations are performed using LAMMPS [57] compiled with QUIP, except for melting point calculations and simulations of liquid and amorphous structures, which are carried out using TurboGAP [32, 33]. In LAMMPS, temperature and pressure are controlled

Table 1. Hyperparameters used during training: r_{hard} (cutoff radius), r_{soft} (r_{hard} - cutoff transition width), $\sigma_{0,r}$ (σ for radial channel), $\sigma_{0,\perp}$ (σ for angular channel), δ (scaling of kernel), l_{max} (number of angular basis functions), α_{max} (number of radial basis functions) and ζ (power of kernel). Sparse method specifies how to choose representative points, with details provided in [46].

	soap_turbo	Two-body
r_{hard}	5 Å	5 Å
r_{soft}	4.5 Å	4.5 Å
$\sigma_{0,r}$	0.5 Å	—
$\sigma_{0,\perp}$	0.5 Å	—
sparse method	CUR	Uniform
δ	1 eV	10 eV
l_{max}	8	—
α_{max}	8	—
ζ	4	—

using Nosé-Hoover [58, 59] thermostat and barostat. In TurboGAP, Berendsen *et al* [60] thermostat and barostat are used for NPT simulations.

The melting temperature of diamond Ge is calculated using the two-phase method [61, 62] with a $8 \times 8 \times 16$ supercell containing 8192 atoms. First, an NPT simulation is performed to equilibrate the system at 600 K and 1 bar for 20 ps, with a timestep of 2 fs and damping constants of 0.2 ps for temperature and 2 ps for pressure. Next, half of the system is fixed while the other half is heated to 1400 K for 20 ps using the NVT ensemble. Finally, the half-crystalline, half-molten system is simulated using the NpH ensemble at 1 bar for 200 ps to observe the coexistence of both liquid and solid phases. The melting temperature is determined from the last 100 ps of the simulation, with the average value taken as the melting temperature for this run. Such simulations are repeated 24 times to obtain the mean melting point.

2.4. Database

All structures included in the training database are listed in table 2. To capture the elastic response of diamond Ge, structures are distorted within a volume range of approximately -35% to 35% of the equilibrium volume. Other crystalline structures including hexagonal diamond (hd), β -Sn, bc8, st12, hexagonal close packed (hcp), body-centered cubic (bcc), and face-centered cubic (fcc) are treated similarly to diamond Ge to demonstrate GAP's ability predicting relative stability of different crystalline structures.

Accurate modeling of atomic vibrations (phonons) is essential for predicting finite-temperature phase stability, thermal expansion, and heat capacity, among other key material properties. There are two methods for first-principles phonon calculations: the finite-displacement supercell approach and density-functional perturbation theory [63], with the former being more advantageous for practical implementation [47]. The training data for phonon calculations are typically obtained from MD trajectories sampled at different temperatures [64] or randomly distorted structures [65]. However, constructing such a training database is computationally expensive due to the large supercells, which contain hundreds of atoms. Instead of relying on the routine method, we employ non-diagonal supercells (NDSCs), as introduced by Lloyd-Williams and Monserrat [66], to generate much smaller cells containing either 2 or 6 atoms, significantly reducing the cost of *ab initio* calculations. A series of NDSCs is generated using the FORTRAN 90 program by Allen and Bartók [67], and the atoms in these cells are perturbed according to a normal distribution with a standard deviation (SD) of 0.05 Å using the rattle function from the Atomic Simulation Environment [68].

A series of monovacancy structures is included in the training database to explore their stability and dynamics. These configurations are derived from a DFT-relaxed structure, with atomic positions perturbed using a normal distribution with a SD of 0.1 Å. Additionally, typical self-interstitial atoms (SIAs) in diamond Ge are incorporated, including $\langle 110 \rangle$ dumbbells and atoms positioned at tetrahedral, hexagonal, and bond-center sites [69]. SIA structures are generated following a similar procedure as for monovacancy structures, but with a larger perturbation SD of 0.15 Å.

To capture the melting process, liquid Ge data are prepared. These liquid structures are sampled by MD simulations with the GAP, which is pre-trained with all the other bulk structures in the training database. A $2 \times 2 \times 2$ supercell of the eight-atom diamond cubic cell (64 atoms) is used for NVT simulations, each running for 20 ps with a 1 fs timestep, at temperatures of 1200 K, 1700 K, 2200 K, and 3000 K. Liquid structures are sampled from the last 10 ps of each simulation, with a range of densities around 5.60 g cm^{-3} considered [70].

Table 2. Details of the training database. The table lists the target physical properties, the corresponding configuration type, the number of configurations (No. config.), the number of atoms of each configuration (No. atoms), the number of training local atomic environments (No. env.), the representative atomic environments (No. rep.) and the regularization parameters for energy (σ_{energy}), force (σ_{force}) and virial stress (σ_{virial}). A dash in the regularization column denotes properties excluded from the training process, and the representative atomic environments are not set for liquid data.

Target property	Configuration type	No. config.	No. atoms	No. env.	No. rep.	σ_{energy} (eV)	σ_{force} (eV/Å)	σ_{virial} (eV)
Short range	dimer	31	2	62	10	0.01	0.4	—
Bulk elastic constants	diamond	47	2	94	20	0.001	0.1	0.04
	bcc	40	1	40	10	0.05	1	0.1
	fcc	22	1	22	10	0.05	1	0.1
	bc8	40	8	320	20	0.05	1	1
	hcp	20	2	40	20	0.01	1	0.1
	hexagonal diamond	20	4	80	10	0.01	1	0.1
	β -Sn	20	2	40	10	0.01	1	0.1
	st12	20	12	240	10	0.01	1	0.1
Bulk phonons	non-diagonal diamond supercell	50	2 / 6	292	100	0.001	0.01	0.1
Bulk monovacancies	diamond	76	63	4788	100	0.001	0.1	—
Self-interstitials	tetrahedral	11	217	2387	100	0.001	0.1	—
	$\langle 110 \rangle$ dumbbell	10	217	2170	100	0.001	0.5	—
	hexagonal	10	217	2170	100	0.01	0.1	—
	bond-center	10	217	2170	100	0.01	0.1	—
Liquid and amorphous	liquid	24	64	1536	—	0.005	0.4	—
	All	451		16 451				

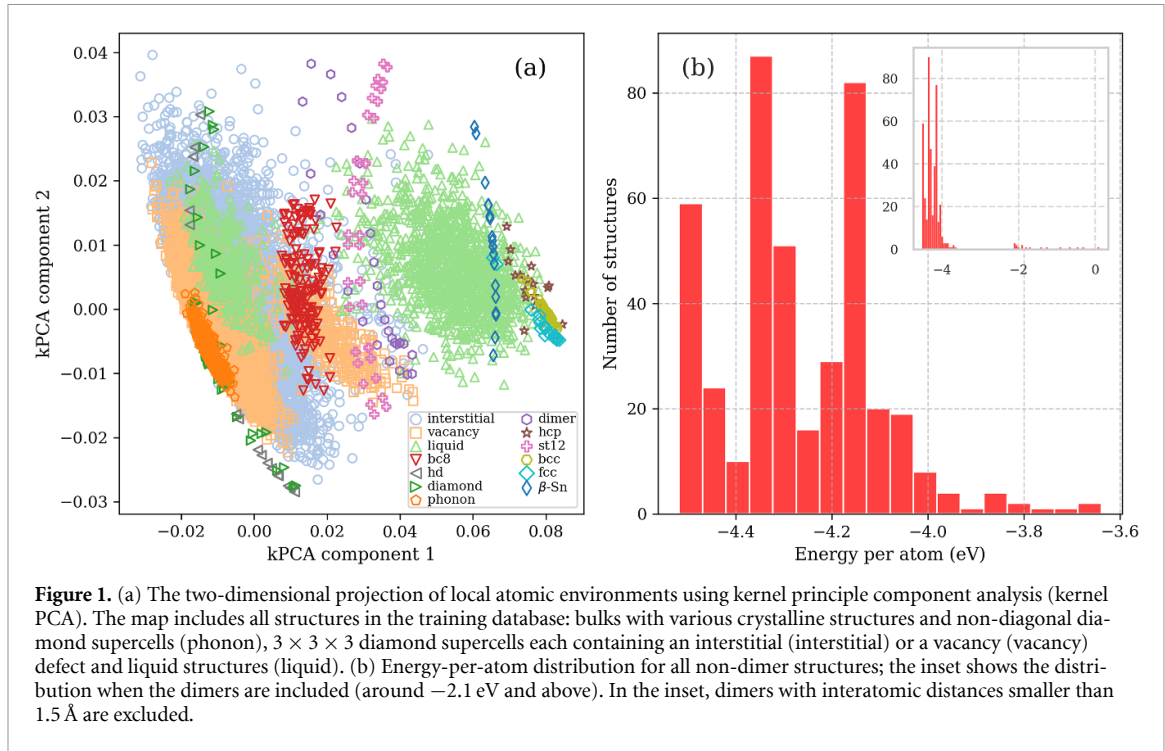


Figure 1. (a) The two-dimensional projection of local atomic environments using kernel principle component analysis (kernel PCA). The map includes all structures in the training database: bulks with various crystalline structures and non-diagonal diamond supercells (phonon), $3 \times 3 \times 3$ diamond supercells each containing an interstitial (interstitial) or a vacancy (vacancy) defect and liquid structures (liquid). (b) Energy-per-atom distribution for all non-dimer structures; the inset shows the distribution when the dimers are included (around -2.1 eV and above). In the inset, dimers with interatomic distances smaller than 1.5 Å are excluded.

Figure 1(a) shows the projected map of the full database, where each point represents an atomic environment and is color-coded by structure type. For this representation, the local descriptors of each atomic environment are computed using the `soap_turbo` descriptor, and kernel principal component analysis (kernel PCA) [71] is used to reduce the high-dimensional descriptor space. The first principal component (PC1) captures the direction of maximum variance in the data, while the second principal component (PC2) captures the next-largest variance and is uncorrelated with PC1. The distribution of local atomic energies in the database is shown in figure 1(b). It shows that the training data primarily concentrated in the energy range of approximately -4.5 to -3.6 eV atom $^{-1}$. The data points around -2.1 eV atom $^{-1}$ and above correspond to dimer configurations (presented in the inset).

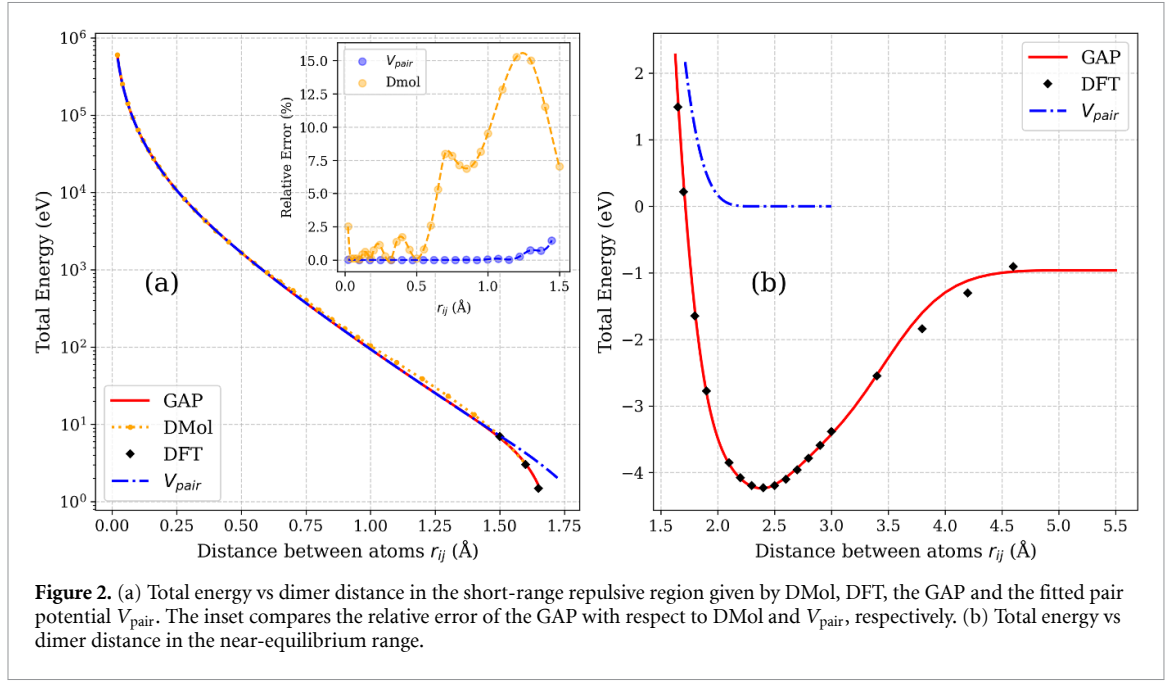
Although the current training database exhibits sufficient spatial and energy diversity for cascade dynamics and the disorder caused by the collision cascade, some limitations exist. Disordered surfaces and planar defects, such as grain boundaries, are not considered here, nor are dislocations covered in the training data set. Hence, this potential is mainly suited for radiation damage studies in pristine microstructure.

Accurately describing short-range repulsion is crucial for radiation damage simulations, particularly during the evolution of collision cascades. The importance of stiffening the MLIP for radiation damage simulations, as well as smoothly joining the repulsive potential to the energies and forces included in the database, has been demonstrated in [25]. In this work, the reference values calculated using DFT, which treats core electrons as frozen with the PAW formalism, show good agreement with those calculated using the DMol approach within all-electron DFT [72] for dimers with atomic distances greater than 1.5 Å. Therefore, only dimer data with atomic distances greater than 1.5 Å, calculated using DFT, are included in the training database. The internuclear repulsion with extremely short dimer distances is captured by incorporating a tabulated pair potential (V_{pair}) corresponding to the first term on the right-hand side of equation (2), which is represented as a screened Coulomb potential and expressed by the following equation:

$$V_{\text{pair}}(r_{ij}) = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{r_{ij}} \phi(r_{ij}/a) f_{\text{cut}}(r_{ij}), \quad (3)$$

where

$$a = \frac{0.46848}{Z_i^{0.23} + Z_j^{0.23}}. \quad (4)$$



The form of the function is similar to the universal Ziegler–Biersack–Littmark(ZBL) potential [73], with a screening function ϕ refitted to DMol data [72]. The fitted screening function is given by

$$\phi(x) = 0.45526 \exp(-1.43007x) + 0.49210 \exp(-0.45871x) + 0.05284 \exp(-0.05284x). \quad (5)$$

In the region where the atomic distances are below 1.5 Å, dimer training data are generated manually, with energies and forces corresponding to the fitted V_{pair} .

In equation (2), the total energy E_{tot} is expressed as the sum of V_{pair} and E_{GAP} , and the smooth transition between V_{pair} and E_{GAP} is achieved via the cutoff function given as

$$f_{\text{cut}}(r) = \begin{cases} 1, & r \leq r_1 \\ 1 - x^3 (6x^2 - 15x + 10), & r_1 < r \leq r_2, \\ 0, & r > r_2, \end{cases} \quad (6)$$

where $x = (r - r_1)/(r_2 - r_1)$, and the transition range is set to $r_1 = 1.5$ Å and $r_2 = 2.3$ Å. The cutoff function is continuous at the boundaries $r = r_1$ and $r = r_2$ up to the second derivative, consistent with the form used in [74]. The effect of the cutoff function is reflected through V_{pair} , where a tabulated format of energies as a function of interatomic distance is used for training. During the training process, V_{pair} is taken as a baseline model [18] and is subtracted from the reference data. In other words, given V_{pair} , the energy difference between V_{pair} and the reference energies in the database is fitted. Then, at the prediction stage, this difference is added back to the baseline.

As shown in the inset plot in figure 2(a), the relative energy error of the GAP with respect to V_{pair} is extremely small, indicating the effectiveness of the manually added dimer data. In contrast, the error relative to the DMol data is higher due to the fitting error of the pair potential. As shown in figure 2(b), the V_{pair} gradually approaches zero, with the GAP accurately capturing the total potential energies in the near-equilibrium region.

3. Results and discussion

In the following sections, we apply our GAP to predict various physical properties, using these properties as benchmarks to evaluate its performance. The evaluation compares the results with available experimental data, as well as with DFT results from this work and from the literature. We highlight the physical properties that our GAP predicts accurately, identify areas where its performance is less satisfactory, and discuss the underlying challenges to further improve the predictive accuracy. In addition to validating performance on physical properties, we compute the root-mean-square errors for energies and forces to evaluate the numerical fitting accuracy. Details are provided in the Supplementary Material (figure S1; tables S1 and S2).

Table 3. Basic properties of diamond Ge: lattice constant a_0 , cohesive energy E_{coh} , elastic constants C_{ij} , bulk modulus B , Young's modulus E , shear modulus G , melting point T_m and vacancy formation energy E_f^v . For comparison, values calculated using the MEAM [13], Tersoff [11] and modified Stillinger–Weber [5] potentials, along with experimental and DFT results are included. The relative errors of the GAP predictions compared to experimental values are listed in parentheses.

	Exp.	DFT	GAP (relative error)	MEAM	Tersoff	Stillinger–Weber
a_0 (Å)	5.658 ^a	5.762	5.760 (1.8%)	5.658 ^f	5.657	5.654
E_{coh} (eV atom ^{−1})	3.85 ^b	3.73	4.04 (4.9%)	3.85 ^f	3.85	3.16
C_{11} (GPa)	128 ^c	105	107 (16.4%)	129 ^f	139	98
C_{12} (GPa)	48 ^c	37	35 (27.1%)	50 ^f	44	49
C_{44} (GPa)	67 ^c	54	48 (28.4%)	67 ^f	67	36
B (GPa)	77 ^d	59	59 (23.4%)	77 ^f	76	66 ^g
E (GPa)	131	108	103 (21.4%)	132	139	80
G (GPa)	54	45	43 (20.4%)	54	58	31
T_m (K)	1210 ^e	—	989 (18.7%)	2200 ^f	2554±38 ^g	1230 ^h
E_f^v (eV)	—	2.24	2.06 (−)	2.31	3.60	3.16

^a Reference [110]

^b Reference [111]

^c Reference [112]

^d Reference [76]

^e Reference [75]

^f Reference [13]

^g Reference [113]

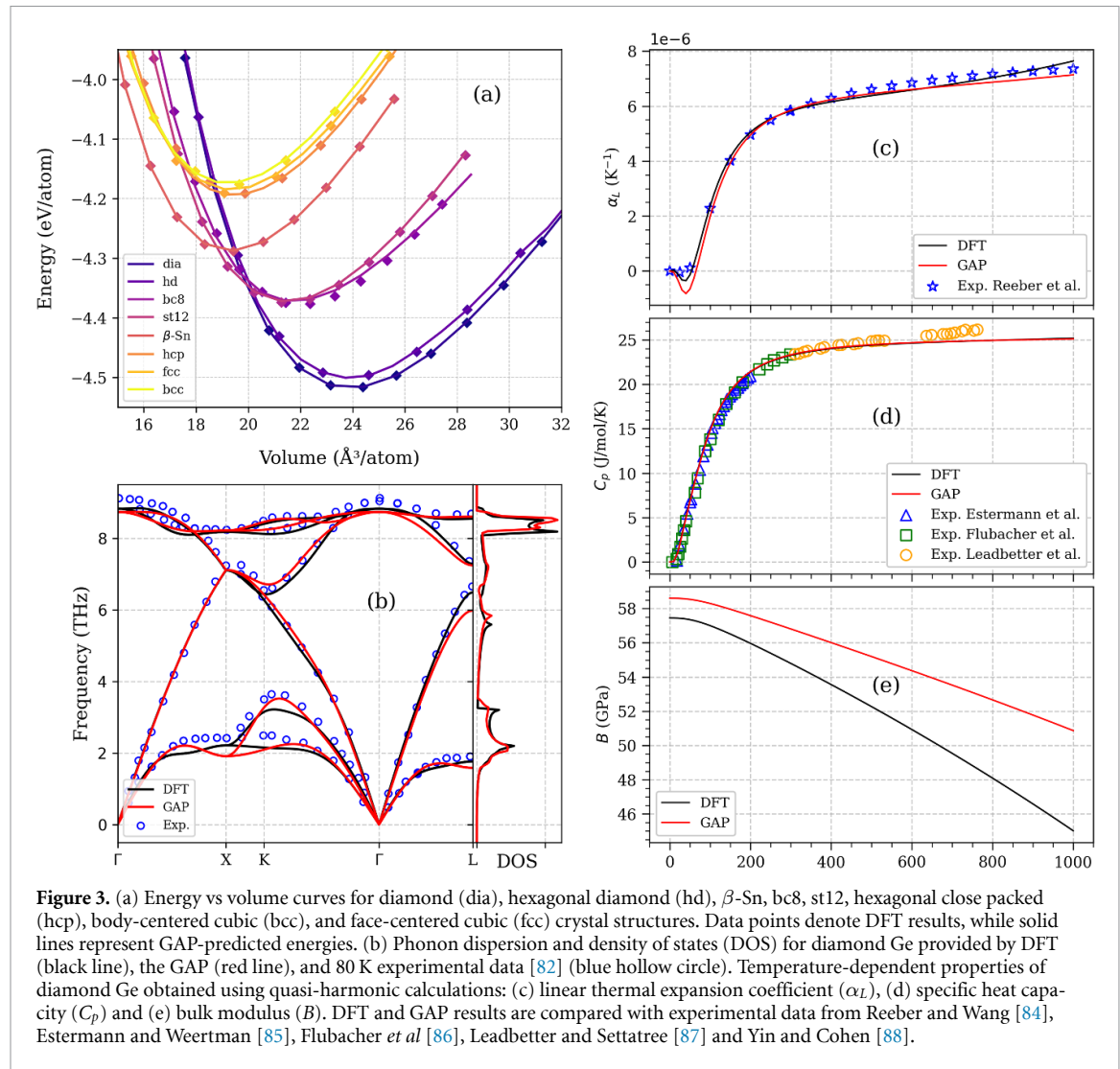
^h Reference [5]

3.1. Bulk crystals

Table 3 presents the basic bulk properties of diamond Ge, including lattice constant a_0 , cohesive energy E_{coh} , melting temperature T_m , monovacancy formation energy E_f^v , elastic constants C_{ij} , and elastic moduli including bulk modulus B , Young's modulus E and shear modulus G . Results are reported from DFT calculations, our GAP model, the empirical MEAM [13], Tersoff [11], and modified Stillinger–Weber [5] potentials, as well as from experimental data. The experimental C_{ij} were measured at 25 °C [75], while the experimental B was measured at 77 K [76]. The Voigt, Reuss, and Hill (VRH) averaging scheme is based on the works of Voigt [77], Reuss [78], and Hill [79]. The relevant equations for these schemes are provided in the supplementary material (section 2). VRH averaging method is used to calculate experimental E and G with experimental C_{ij} values, as well as to determine the elastic moduli for DFT, GAP, and the empirical potentials.

For GAP-predicted results, relative errors with respect to the corresponding experimental values are reported. Compared to our DFT results, our GAP closely reproduces a_0 and B with near-DFT accuracy, with the highest relative error (11.1%) occurring for C_{44} . However, notable discrepancies remain when compared to experimental values, particularly in elastic constants and bulk moduli, with the highest relative error of 28.4% observed in the prediction of C_{44} . In contrast, empirical potentials, especially the MEAM potential, provide a nearly perfect match to experimental results. This is because our GAP is constructed by fitting DFT reference data, and DFT with the GGA exchange–correlation functional tends to overestimate a_0 and underestimate C_{ij} [80]. On the other hand, the MEAM potential was developed by fitting the experimental values directly [13]. The cohesive energy E_{coh} predicted by the GAP is 4.9% higher than the experimental result. For the same reason, the empirical MEAM and Tersoff potentials tend to show better agreement with the experimental value for cohesive energy. To calculate T_m with the GAP, we use a two-phase coexistence method [61, 62], the details of which are provided in the method section. Although the GAP-predicted T_m is slightly lower than the experimental value, it is significantly more accurate compared to the values calculated with MEAM [13] and Tersoff [11] potentials. The discrepancy between the GAP-predicted T_m and the experimental value is attributed to the inherent limitations of DFT, as also observed in [10, 81]. The T_m calculated using the modified Stillinger–Weber [5] potential shows the best agreement with the experimental value, as the modification was designed to reproduce the melting point. The E_f^v is also reproduced by our GAP, matching both our DFT result and the reported DFT value of 2.19 eV [36], while empirical potentials, especially the Tersoff potential, overestimate it.

The energy–volume curves for various crystal structures included in the training database are illustrated in figure 3(a), with energies and volumes normalized per atom by scaling with the number of atoms in the respective unit cells. The GAP nearly perfectly reproduces the DFT results, capturing the



relative stability of different crystal structures and the minimum potential energy with the corresponding volume of each structure. The differences are noticeable only for bc8 lattices, with a volume range of approximately $21\text{--}25 \text{\AA}^3 \text{ atom}^{-1}$, and for hexagonal diamond (hd) structures, at volumes exceeding roughly $31 \text{\AA}^3 \text{ atom}^{-1}$.

Figure 3(b) illustrates the phonon dispersion and density of states (DOS) of diamond Ge, comparing results obtained using our GAP and DFT with experimental data [82]. Considering the significant influence of lattice constants on lattice dynamical properties [83], the lattice constant of natural Ge at 80 K ($5.6524 \text{\AA}$) is chosen for calculating phonon dispersion curves, allowing for an easy comparison with experimental values. Phonopy [47] is used for all phonon calculations, with DFT computations performed using a $5 \times 5 \times 5$ primitive cell and MD simulations conducted with a $6 \times 6 \times 6$ primitive cell. While our GAP shows generally good agreement with DFT in phonon dispersion, noticeable discrepancies exist near X, K, and L points. The observed discrepancies between DFT and experimental results at high frequencies are attributed to the choice of exchange-correlation functional [83]. Notably, DOS is almost perfectly reproduced by our GAP, with minor deviations at high frequencies. In addition, we find that incorporating additional crystalline phases such as st12 structures in the training database may compromise the accuracy of phonon dispersion predictions.

The thermal and mechanical properties, including linear thermal expansion coefficient (α_L), specific heat capacity (C_p) and bulk modulus B , as a function of temperature for diamond Ge are presented in figures 3(c)–(e). The calculations are all performed within the quasiharmonic approximation implemented in the phonopy code [47]. As shown in figure 3(c), the experimental α_L values [84] are nearly reproduced by DFT and the GAP. Although both DFT and the GAP present negative α_L consistent with experimental data, noticeable differences emerge at temperatures below 200 K and above 700 K. In figure 3(d), the experimental values of C_p [85–87] as a function of temperature are well captured by

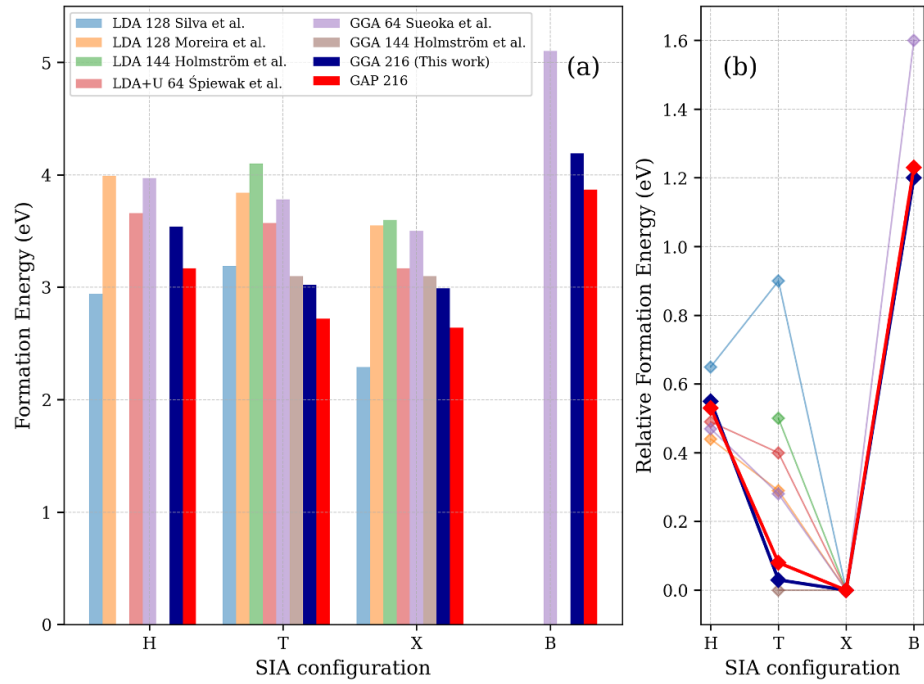


Figure 4. (a) Formation energies of self-interstitials including hexagonal (H), tetragonal (T), $\langle 110 \rangle$ dumbbell (X) and bond-center (B) configurations. (b) Relative formation energies with respect to that of the X configuration. The values are provided by our GAP, DFT from this work, and DFT from Silva *et al* [106], Moreira *et al* [107], Holmström *et al* [89], Śpiewak *et al* [108] and Sueoka *et al* [109]. The legend specifies the exchange-correlation functional (local density approximation (LDA) and generalized gradient approximation (GGA)) and the number of atoms in the system for each DFT calculation.

both DFT and the GAP. While the calculated values are slightly lower than the experimental data at temperatures above 600 K, the DFT values are completely reproduced by our GAP. Figure 3(e) shows the calculated B with varying temperatures, along with the experimental value at 77 K [88]. At 77 K, B calculated by both DFT and the GAP shows a significant deviation from the experimental value. Moreover, compared to DFT results, GAP-predicted B shows a weaker temperature dependence.

3.2. Point defects

Several point defects are included in the training database, and our GAP is expected to predict their formation and migration energies with near-DFT accuracy. The formation energy of a single vacancy is given in table 3. The GAP slightly underestimates this value, with a deviation of 8.0% from our DFT result. The migration energy barriers between first nearest neighbor (1NN) and second nearest neighbor (2NN) are calculated using the NEB [54] method for both DFT and MD simulation. DFT calculations are converged to a maximum force criterion of $0.03 \text{ eV } \text{\AA}^{-1}$, whereas GAP calculations are converged to $0.001 \text{ eV } \text{\AA}^{-1}$. The migration energy barriers between 1NN and 2NN are 0.22 and 1.77 eV as calculated by DFT, and 0.21 and 1.71 eV by our GAP, respectively.

Point defects in the form of SIAs in diamond Ge have been extensively studied, particularly for the hexagonal (H), tetragonal (T), $\langle 110 \rangle$ dumbbell (X) and bond-center (B) configurations. Structures with these SIAs are included in our training database, and their formation energies are calculated using the GAP and DFT with a $3 \times 3 \times 3$ conventional cell, in comparison with various DFT results from the literature. As shown in figure 4(a), DFT calculations with different cell sizes and exchange-correlation functionals yield different formation energies for each SIA. For T and X configurations, the DFT results from this work agree well with those from [89]. On average, GAP-predicted formation energies are 0.31 eV lower than DFT results from this work. Figure 4(b) shows the formation energy of each SIA relative to that of X configuration. It is found that the X configuration is identified as the most stable across all DFT results, including those from our GAP. Additionally, the GAP reproduces relative formation energies with excellent agreement to DFT results from this work.

3.3. Liquid and amorphous

To characterize the liquid structure of Ge using our GAP, we perform MD simulations with the TurboGAP code [32, 33]. In TurboGAP, Berendsen [60] thermostat is used with a damping constant of 100 fs. A $10 \times 10 \times 10$ supercell of the conventional unit cell (8000 atoms) is heated to 1300 K for

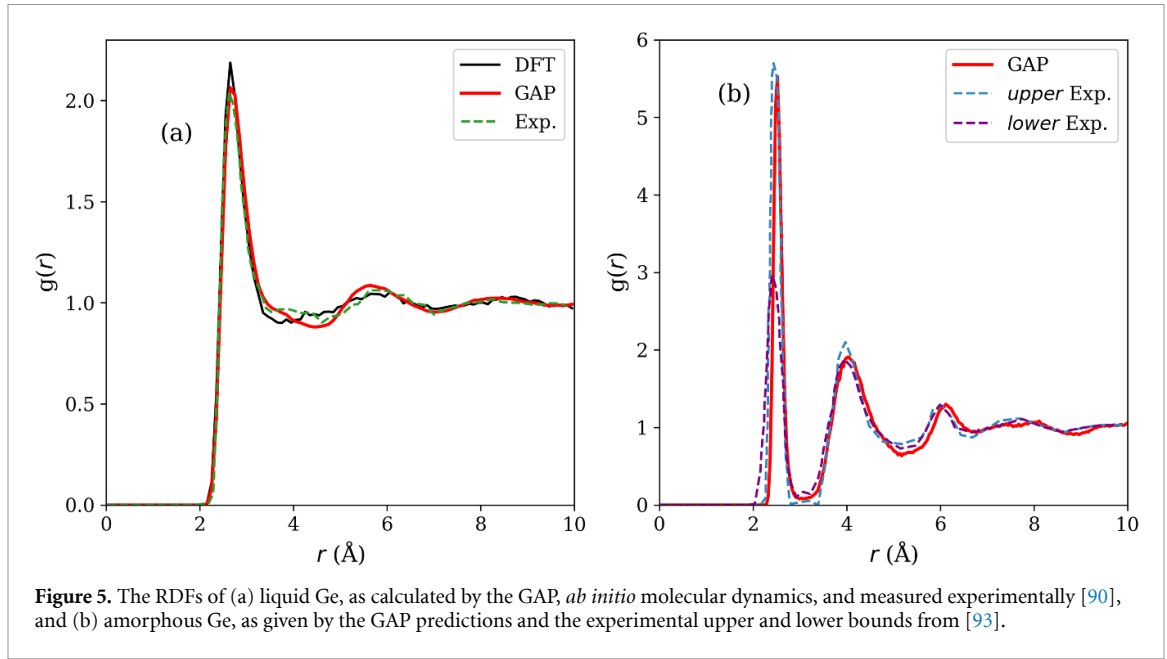


Figure 5. The RDFs of (a) liquid Ge, as calculated by the GAP, *ab initio* molecular dynamics, and measured experimentally [90], and (b) amorphous Ge, as given by the GAP predictions and the experimental upper and lower bounds from [93].

30 ps with a timestep of 1 fs. The structural data from the last 500 steps are extracted and averaged for plotting the radial distribution function (RDF). Reference DFT results are obtained by performing *ab initio* MD (AIMD) simulation of a $2 \times 2 \times 2$ cubic-diamond Ge supercell, running 12 ps (1300 K, 1 fs timestep). The simulation employs a 350 eV plane-wave energy cutoff and uses the Γ point for the Brillouin zone integration, with the temperature controlled by Nosé–Hoover thermostat, and the structural data are collected from the last 5000 frames. The Ge density ρ is set to 5.50 g cm^{-3} , close to its experimental density at 1000°C [90]. The resulting structural quantities are shown in figure 5(a), alongside the experimental data from [90] for comparison. The GAP-predicted RDF shows excellent agreement with both DFT and experimental data, accurately capturing the 1NN peak near $2.6 \text{ \AA}$ and its height. As the distance r increases up to $10 \text{ \AA}$, the RDF values gradually approach 1. The coordination number (CN) of the liquid is obtained by evaluating the integral $\text{CN}(r_c) = 4\pi\rho \int_0^{r_c} g(r) r^2 dr$, where the cutoff radius $r_c = 3.5 \text{ \AA}$ corresponds to the first minimum of the RDF. This yields a CN of 7.6, which is close to the experimental value of 8.0 ± 0.2 reported for the first-shell CN of liquid Ge in [90].

The amorphous phase can form under irradiation, primarily due to the spontaneous local collapse of the crystal structure and the loss of long-range order resulting from a high concentration of defects, while retaining some short-range order [91, 92]. It can also be produced by rapid quenching, a method commonly employed in simulations. The structure of amorphous Ge is modeled by performing quenching simulations on a 216-atom liquid Ge sample. The system is first equilibrated at 1800 K for 100 ps with a timestep of 1 fs, then cooled from 1800 K to 300 K at a quenching rate of 10^{12} K s^{-1} for 1500 ps (1.5×10^6 steps), maintaining the pressure of 1 bar, using the TurboGAP code. The RDF of the amorphous structure is illustrated in figure 5(b), in comparison with experimental lower and upper bounds from [93]. DFT calculations are not feasible in this case due to the slow quenching rate. Unlike the RDF of the liquid structure, the RDF of the amorphous phase typically exhibits structural order extending at least up to the 2NN peak [92]. Figure 5(b) shows that the amorphous Ge exhibits a sharp first neighbor peak around $2.5 \text{ \AA}$, a second peak near $4 \text{ \AA}$, and a broad third peak at $6 \text{ \AA}$, all within the upper and lower experimental bounds. Although the amorphous Ge structures are not explicitly included in the training database, the GAP nearly perfectly reproduces the experimental RDF of amorphous phase, demonstrating its extrapolative capability.

3.4. Repulsive potential

To evaluate the performance of the GAP for cascade simulations, we perform quasi-static drag (QSD) simulations, in which a single atom is gradually displaced along a specified direction from its original lattice site to a defect site, while all other atoms remain fixed. The total energy difference is the energy of the defective system minus that of the perfect crystal. This approach is similar to the sudden approximation method, as described in [94]. Since the chosen atom moves along specific directions, this method is only applicable in simple cases where the defect is formed along a straight path. Furthermore, as the atom is moved through the lattice, which involves many-body interactions, this process also serves as an

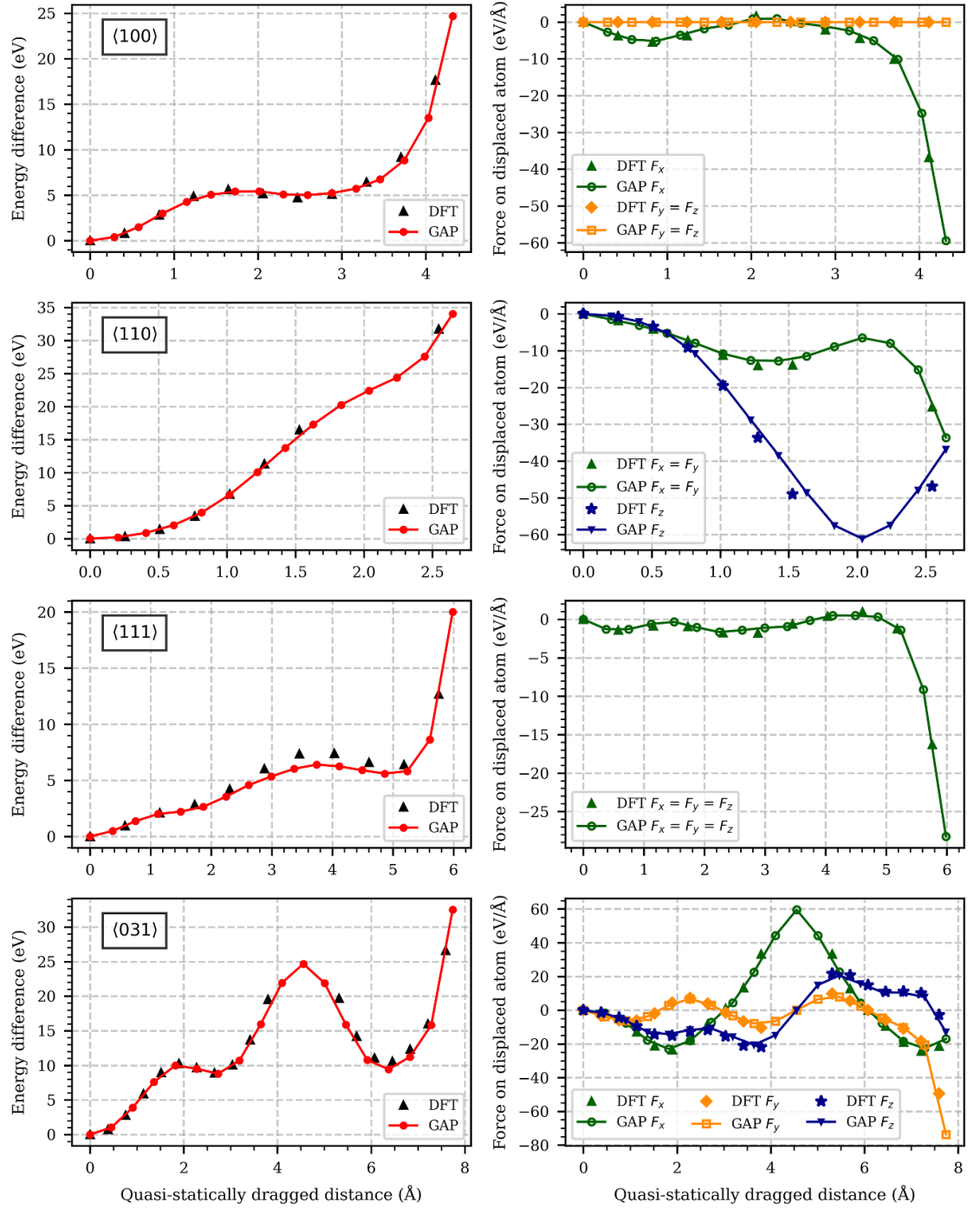


Figure 6. Total energy difference (left) and the corresponding force components (right) for the stepwise movement of one atom along $\langle 100 \rangle$, $\langle 110 \rangle$, open $\langle 111 \rangle$ and $\langle 031 \rangle$ directions in diamond Ge.

effective test of the short-range many-body behavior [95]. We use a $2 \times 2 \times 2$ supercell of the diamond cubic cell (64 atoms) to run QSD simulations with both DFT and the GAP. Additionally, in the QSD simulations with DFT, calculations are restricted to interatomic distances ≥ 1.5 Å, where DFT remains accurate. The total energy difference and force components on the displaced atom along four different crystal directions are presented in figure 6. The simulation results from DFT and our GAP agree quite well, except for notable discrepancies in the total energy difference along the open $\langle 111 \rangle$ direction. Despite not being trained on these configurations, the GAP closely matches DFT-calculated total energy differences and forces on the displaced atom. This can be considered a good validation of GAP's performance in cascade simulations.

Table 4. The threshold displacement energies along different directions calculated by DFT, our GAP and the MEAM potential. O and C denote the open and closed [111] directions, respectively. The bold values represent those relatively closer to the DFT results.

	DFT (eV)	GAP (eV)	MEAM (eV)
[001]	18 ^a , 19.5 ^b , 18 ^c	11.0	19.5
[101]	28.5 ^a	14.9	24.7
[111] (O)	9.5 ^a , 11.5 ^a , 15 ^d	7.6	19.5
[111] (C)	9.5 ^a , 12.5 ^b	6.6	17.9

^a Reference [99]

^b Reference [89]

^c Reference [114]

^d Reference [101]

The threshold displacement energy (TDE) is the minimum kinetic energy required to generate a single Frenkel pair (FP) in a material, which is a critical parameter for investigating radiation damage. The TDE value is highly dependent on the initial velocity direction of the PKA and is determined by incrementally increasing the kinetic energy imparted to the PKA until it is permanently displaced from its lattice site to a new stable position, as described in [96]. The initial velocity directions are uniformly sampled over the symmetry-reduced sphere using a method inspired by [97]. The MD simulations are carried out on a $9 \times 9 \times 9$ supercell of the diamond cubic structure (5832 atoms), which is sufficiently large to prevent interactions between the displaced atom and the supercell boundaries. The simulations are performed at 0 K for a duration of 5 ps using an adaptive timestep up to 0.002 ps, with a maximum allowed kinetic energy change of 10 eV for an atom in one timestep. The kinetic energy is increased in steps smaller than 3 eV, and the formation of a FP is detected using the Wigner-Seitz defect analysis as implemented in OVITO [98]. For the calculations with the MEAM potential, it is splined to the ZBL potential [73] using the standardized method implemented in LAMMPS [57]. In this approach, the full ZBL potential is applied for interatomic distances up to around 1 Å, followed by a linear switching to the MEAM potential until about 2 Å. The original MEAM is thus modified only at distances which are shorter than the spacing between the split dumbbell interstitial atoms, hence neither equilibrium nor defect properties are affected.

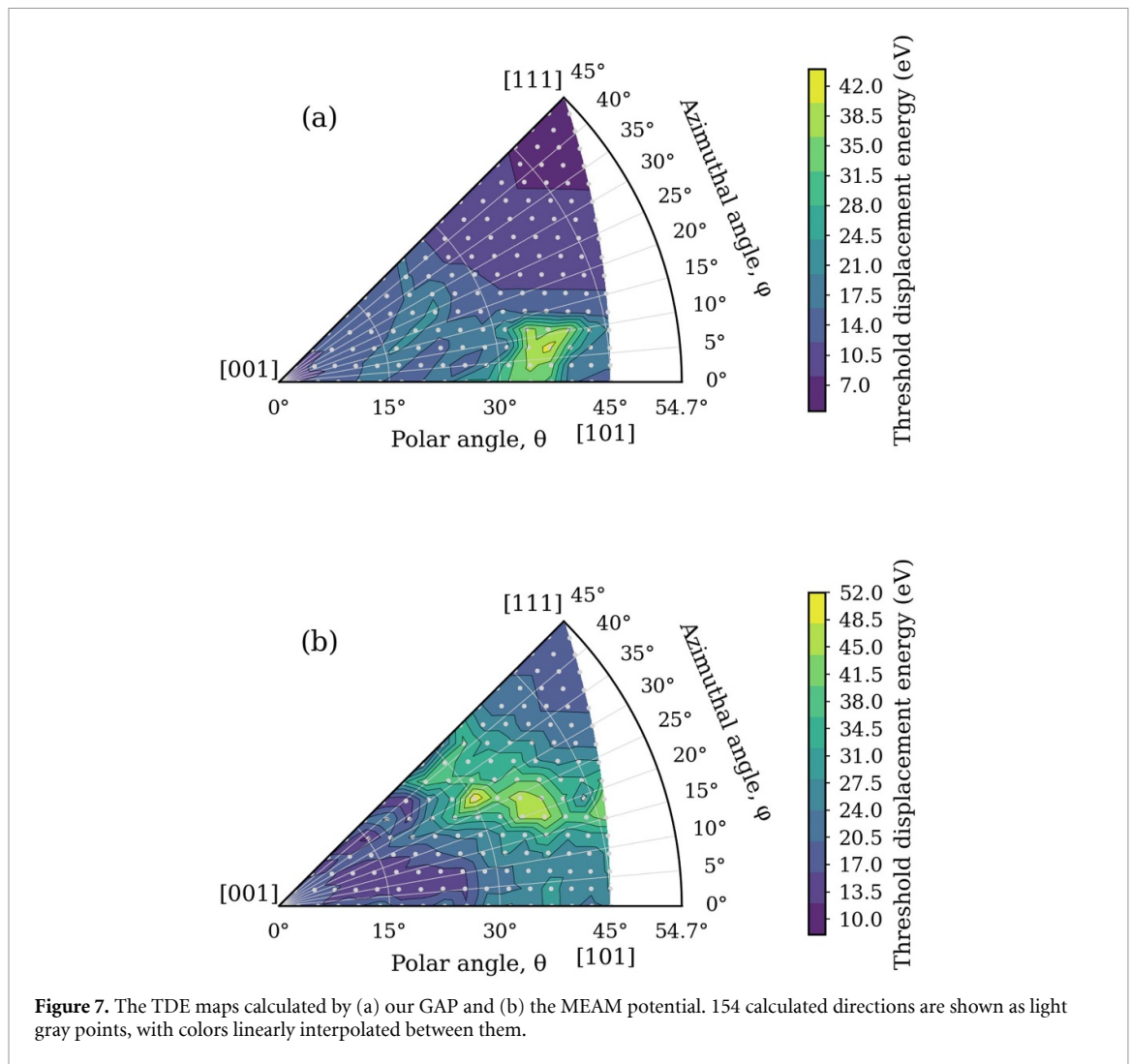
Table 4 presents the TDE values for low-index directions calculated using our GAP and the MEAM potential, compared with the AIMD values reported in the literature. The GAP-predicted TDE values along the open and closed [111] directions are 7.6 and 6.6 eV, respectively, which are in good agreement with the 9.5 eV AIMD results reported in [99]. However, GAP underestimates the TDE in the [001] and [101] directions compared to AIMD. In contrast, the MEAM potential shows good agreement for the [001] and [101] directions but overestimates the TDE for the [111] directions.

The angular TDE maps obtained using our GAP and the MEAM potential [13] are shown in figures 7(a) and (b), respectively. In figure 7, the 154 sampled directions are indicated by light gray data points, while the colored regions represent values obtained through linear interpolation. The global minimum TDE obtained by GAP is approximately 7 eV, which can be compared to the value of 11.5 eV reported from AIMD simulations in [89]. Experimental measurements indicate that the minimum TDE over all lattice directions lies in the range of approximately 15–30 eV [100–103]. Despite the good agreement between GAP and AIMD, both methods underestimate the experimental values. The global minimum for the MEAM potential is 10 eV. Another notable feature of the TDE angular map is a localized directional region with high resistance to defect formation in the GAP prediction, corresponding to regions where the TDE values are elevated. In figure 7, the region with $30^\circ < \theta < 45^\circ$ and $0^\circ < \phi < 15^\circ$ exhibits higher TDE values compared to the other sampled directions. In this regard, the GAP prediction differs considerably from that of the MEAM potential, which shows a broader high-TDE region spanning $25^\circ < \theta < 45^\circ$ and $15^\circ < \phi < 45^\circ$.

The global average TDE (T_{ave}) is obtained by numerically integrating the interpolated energy values over the symmetry-reduced portion of the unit sphere, using the following equation:

$$T_{ave} = \frac{\iint T(\theta, \varphi) \sin \theta d\theta d\varphi}{\iint \sin \theta d\theta d\varphi}, \quad (7)$$

where $T(\theta, \varphi)$ denotes a TDE value along the direction defined by the polar angle θ and the azimuthal angle φ . The T_{ave} calculated with the GAP is 14.7 eV, slightly smaller than the experimental values, which



range from 18 [101] to 30 eV [103]. For comparison, AIMD simulations reported an average TDE of 23 eV based on 80 random directions [89]. Overall, considering both the AIMD and experimental data, GAP demonstrates good performance in predicting the TDE in Ge. The T_{ave} for the MEAM potential is 25.7 eV.

3.5. Cascade simulations

To investigate the potential's performance in radiation damage simulations, We perform cascade simulations with a PKA energy of 2 keV using both our GAP and the MEAM empirical potential [13]. The system is first equilibrated under the NVT ensemble and then under the NPT ensemble until the temperature stabilizes at approximately 300 K and the pressure fluctuates around 0 bar. A cascade simulation is performed in a supercell comprising 85 184 atoms, with a 5.76 Å-thick boundary region acting as a heat sink. The heat sink temperature is maintained at 300 K using Nosé–Hoover thermostat. The cascade simulation runs for 30 ps with an adaptive timestep. The maximum allowed timestep is set to 0.003 ps, and the maximum kinetic energy change of any atom is limited to 30 eV per timestep. Electronic stopping [104] is applied as a non-local friction force to the atoms with kinetic energies above 10 eV.

Variation of the temperature and increased potential energy are shown in figure 8. After the ballistic stage, the cascade enters the heat spike phase, where the damaged volume ceases to grow and a disordered, underdense, liquid-like core forms in the cascade region. Subsequently, the cascade cools down and the system gradually approaches equilibrium. Our GAP shows the expected behavior [5, 27, 105] in

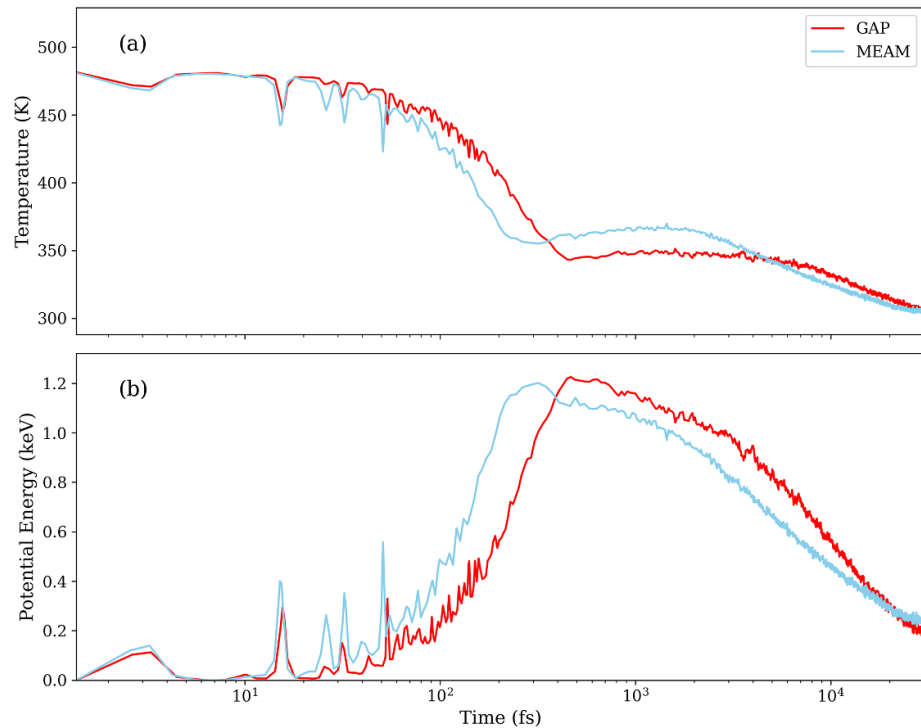


Figure 8. Time evolution of (a) temperature and (b) increased potential energy for a 2 keV PKA collision cascade simulated using our GAP and the MEAM potential in Ge.

cascade simulations, demonstrating its stability and reliability in capturing the dynamics of radiation-induced displacements and their subsequent evolution.

To assess the reliability of the potential's predictions, we evaluate the uncertainty in the predicted energies using a snapshot taken roughly 1 ps after the PKA is introduced. This time frame corresponds to the heat spike stage of the cascade evolution. The QUIP package [56] is used to perform the uncertainty evaluation. Figure 9(a) shows the prediction uncertainty across the simulation cell. Atoms with uncertainties above $0.04 \text{ eV atom}^{-1}$ are displayed with an increased radius for clarity; the black circle marks the PKA position, and the arrow indicates its initial velocity direction. It shows that atoms near the PKA and along the velocity direction exhibit higher prediction uncertainty. Figure 9(b) presents the prediction uncertainty as a function of local energy, with points color-coded by relative uncertainty (uncertainty divided by local energy). The minimum and maximum per-atom energies in the training database, excluding dimer structures, are indicated by vertical dotted lines. As shown, nearly all atomic energies in the snapshot fall within the energy interval covered by the training data. Figure 9(c) displays the descriptor-space projection of the atomic environments in the training database overlaid with those from the snapshot, color-coded by their predicted uncertainty. A good overlap between the atomic environments encountered during the displacement cascades and those present in the database is observed. This provides confidence that the trained GAP model yields reliable predictions for cascade simulations in Ge. In figure 10, we present snapshots of the cascade simulation, illustrating its generation, interactions, and evolution from 0.1 to 30 ps. Atoms with atomic potential energy above -4 eV atom^{-1} are plotted with enlarged radii and color-coded according to their kinetic energies.

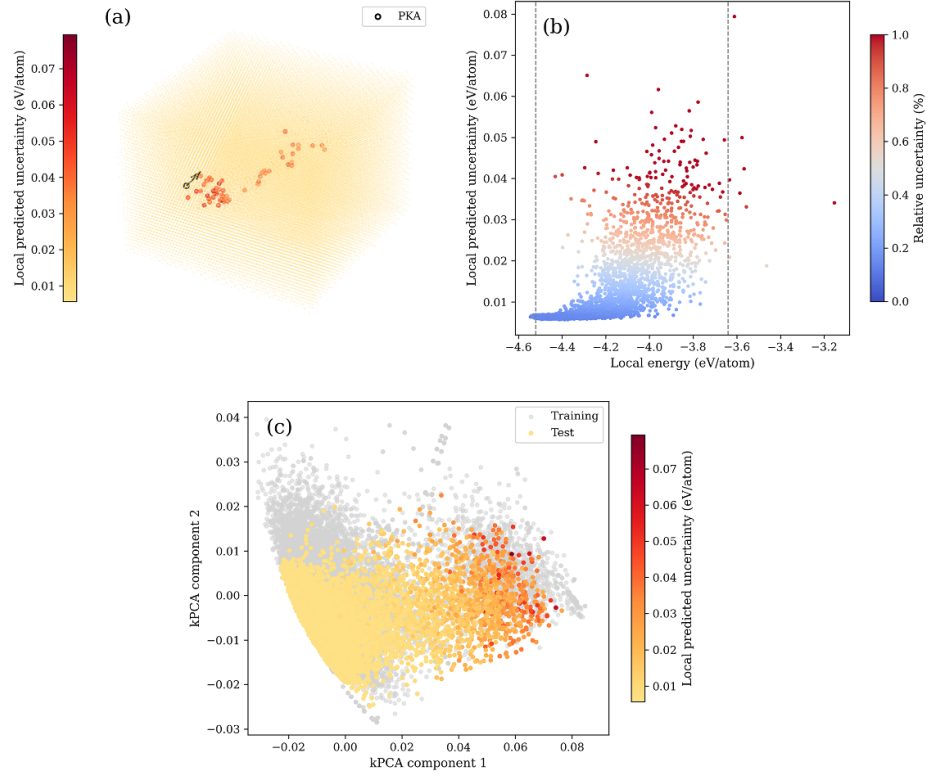


Figure 9. Uncertainty analysis for a snapshot from cascade simulation derived from a 2 keV PKA, taken at the time frame of 1 ps. (a) Local predicted uncertainty across the simulation cell, with the black circle indicating the PKA and the arrow showing its velocity direction. (b) Local predicted uncertainty vs local energy, with dots color-coded by the uncertainty relative to the corresponding energy. The two grey dotted lines indicate the minimum and maximum energies in the training database, excluding dimers. (c) Kernel PCA projection of the descriptor space for both the training database and this snapshot.

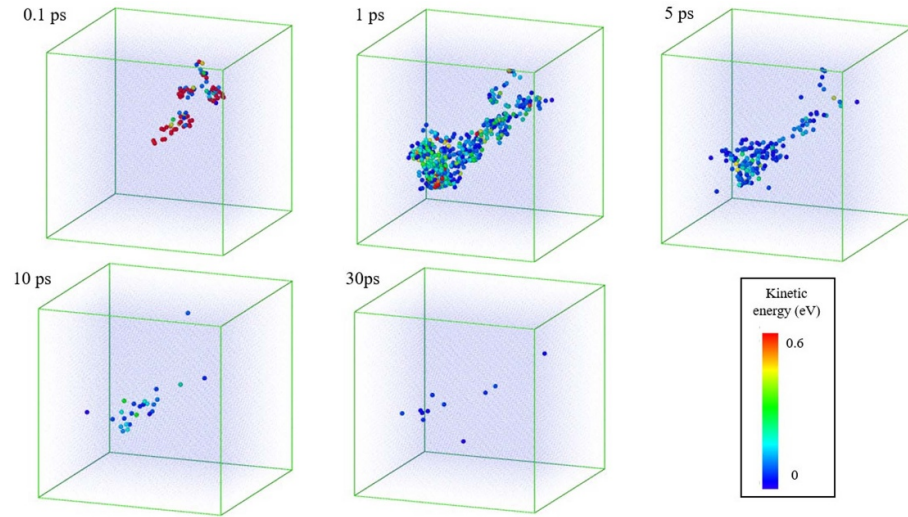


Figure 10. Snapshots of primary damage induced by a 2 keV PKA in Ge, simulated using our GAP. Atoms with potential energy above -4 eV atom^{-1} are shown with enlarged radii and are color-coded by their kinetic energies.

4. Conclusion

In the preceding sections, we show how a machine-learning potential GAP, trained on a relatively small database, effectively captures a wide range of Ge properties with near-DFT accuracy. Focusing specifically on radiation damage-related properties, the repulsive term is incorporated to capture short-range many-body behaviors. This potential can describe a broad range of fundamental physical properties, including elastic, thermal (notably the melting point), and vibrational properties, as well as the energetics of point

defects. The TDE values are predicted more accurately, particularly along the open and closed [111] directions, compared to the MEAM potential. The low prediction uncertainty evaluated in one cascade simulation snapshot further confirms the reliability of this potential. Additionally, large-scale MD simulations carried out with our GAP via the TurboGAP code exhibit excellent computational efficiency, as shown in Supplementary Material, section 4. Overall, with its modest training data requirements, precise handling of short-range repulsion, and efficiency in large-scale simulations, this GAP facilitates future studies of radiation damage in Ge.

Data availability statement

The training database and the potential files are available at <https://doi.org/10.5281/zenodo.17305040>. All codes used in this work are available in the public GitHub repository *ilearn* at <https://github.com/Ruoyan0103/ilearn>.

Acknowledgments

The authors wish to thank Miguel Caro, Richard Jana, Rina Ibragimova, Jarno Laakso and Pascal Henkel for useful discussions, and acknowledge the computational resources provided by Aalto Science-IT and CSC – IT Center for Science, Finland. This work was funded by the European Union (ERC, MUST, 101077454). Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. Computing resources from CSC—IT Center for Science are Gratefully acknowledged.

Author contributions

Ruoyan Jin carried out the simulations and wrote the manuscript. Both Ali Hamedani and Andrea E. Sand contributed to the final version of the manuscript. Andrea E. Sand supervised the project.

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